# Report on Activity Based Learning

Teacher's Assessment Examination (TAE-1)

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| Session | 2025 -26 | Semester | II |
| Program | B. Tech. | Course | Chemistry For Engineering |
| Course Code | N-BSC103T | Section | C |
| USN No. | CS25128 | | |
| Name of the Student | Aditi Karan | | |
| Title of the Activity | Study of Biodegradability Test of various polymers by using common chemicals. | | |
| Signature of the Student | | Date of Display | **17/03/2026** |
| | | Date of Submission | |

## Evaluation Report

| Parameters | Marks Obtained | Maximum Marks |
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| Performance and Submission of Report | | 4 |
| On time submission | | 1 |
| **Total** | | **5** |

| Content | Details (Expected explanations) |
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| **A) Introduction** | **The Problem with Conventional Plastics**<br>• Environmental Impact: Persistent synthetic plastics have caused global concern due to rising environmental pollution.<br>• Resistance to Decay: Common plastics like polyethylene (PE) and polypropylene (PP) are highly resistant to degradation. They can remain in ecosystems for hundreds of years without breaking down.<br>**The Rise of Biodegradable Alternatives**<br>• Interest in Bioplastics: There is growing interest in materials designed to break down more easily under environmental conditions, such as: |

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| | I. Polylactic acid (PLA).<br>II. Polycaprolactone (PCL).<br>III. Starch-based plastics.<br>• True Biodegradation: This process involves microbial activity that breaks polymers down into water, carbon dioxide, and biomass.<br>**The Role of Chemical Testing**<br>• Simulated Testing: While true biodegradation testing is time-consuming and requires specific soil or composting conditions, chemical degradation serves as an accessible alternative.<br>• Methodology: By using common household chemicals like acids (vinegar), bases (baking soda), and oxidizers (hydrogen peroxide), students can simulate environmental exposure.<br>• Objective: By monitoring physical changes (cracks, softening) and weight loss over time, the activity assesses how susceptible different polymers are to breakdown, providing insight into their potential environmental footprint. |
| B) **Procedure** | **1. Preparation of Polymer Samples**<br>• Each polymer sample (such as PLA, PCL, or Polyethylene) must be cut into **uniform strips**, specifically measuring 4 cm x 1 cm.<br>• Each individual piece must be labeled for identification (e.g., PLA-1, PCL-1) to ensure tracking throughout the experiment.<br>**2. Preparation of Testing Solutions**<br>• Acidic Solution: Use pure vinegar with no dilution (approx. 5% acetic acid).<br>• Alkaline Solution: Dissolve 1 tablespoon of baking soda in 250 ml of warm water.<br>• Oxidative Solution: Use 3% Hydrogen peroxide, typically sourced from a pharmacy.<br>• Control Solution: Use tap water or distilled water to serve as a baseline for comparison.<br>**3. Soaking the Samples**<br>• Place one strip of each polymer type into each of |

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| | the four solutions prepared above.<br>• The jars or beakers should be covered loosely to prevent the solutions from evaporating.<br>• Store the containers at room temperature and ensure they are kept away from direct sunlight.<br>**4. Monitoring and Analysis**<br>• Timeline: Samples must be checked every 2–3 days for a total duration of 2–3 weeks.<br>• Observation Criteria: At every interval, students must measure the weight of the samples and look for the following physical changes:<br>I. Surface changes: Evidence of cracks, softening, or discoloration.<br>II. Flexibility: Whether the material has become brittle or soft.<br>III. Size: Any visible reduction in the dimensions of the strip.<br>• **Final Step:** Analyze the gathered data to rank the polymers based on their reactivity to the different chemical environments. |
| C) **Result and discussion** | **Results: What happened to the samples?**<br>The experiment showed how different materials reacted when left in different liquids for 2–3 weeks:<br>• Starch-based plastics: These were the most sensitive. They softened very quickly and broke down fast, especially in the baking soda (base) and vinegar (acid) solutions.<br>• PCL: This material showed a "moderate" or medium amount of change across all the different liquids.<br>• PLA: This showed some signs of breaking down, particularly in the hydrogen peroxide and vinegar, but it happened more slowly than the starch plastic.<br>• Polyethylene (PE): This is the plastic used in regular grocery bags. It showed no change at all in any of the liquids.<br>**Discussion: What do these results mean?**<br>The results help us understand how plastic waste affects the planet:<br>• Bioplastics are more "reactive": Because materials |

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| | like PLA and starch-based plastics showed clear signs of softening and cracking, we know they are more likely to break down (biodegrade) when exposed to nature.<br>• Since the polyethylene (PE) did not change at all, it proves that regular plastics are very resistant to chemicals. This is why they can stay in the environment for hundreds of years without disappearing.<br>• Why this matters: The activity proves that using biodegradable alternatives is better for the environment because they are designed to break down much faster than traditional plastics. |
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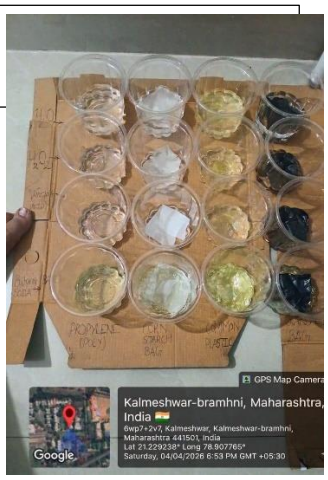
| D) **Proofs** | Geotag images of the activity performed |
|  <br>Date: 04/04/26 | <br>Date: 08/04/26 |
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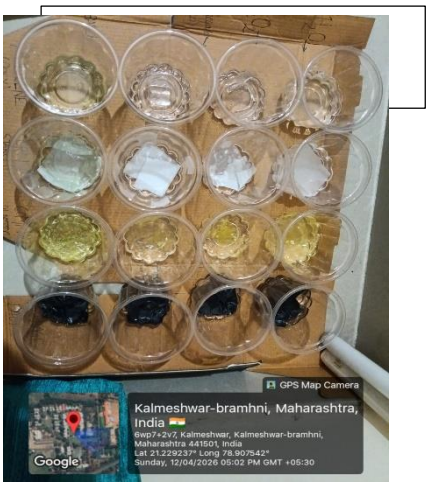
|   | |
| Date: 12/04/16 | Date: 16/04/26 |

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